BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 046 · Algorithmic Thinking

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is decomposition?

Q2. Which is an OS file-management task?
A. Adding two scores
B. Renaming a folder
C. Routing a packet
D. Decoding an instruction

Q3. “Lossless compression deliberately removes arbitrary source-code characters.” Correct this. Explain why.

Q4. A computer needs maintenance and data protection. “Utilities are only apps for writing essays.” Correct this. What do utilities do?

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I Do · Read new knowledge

Key vocabulary

Algorithm — Finite sequence of clear instructions to solve a problem.

Algorithmic thinking — Creating an ordered, logical method for a task.

Sequence — Instructions carried out in their specified order.

Input — Data supplied to an algorithm.

Processing — Actions or calculations on data.

Output — Result provided by an algorithm.

Dry run — Manually stepping through an algorithm with example values.

New learning

What is an algorithm?: An algorithm is a finite sequence of clear, ordered instructions that solves a problem. The steps must be sufficiently precise and unambiguous to follow.

Find inputs, actions and result: Read the task and identify what information is supplied, what calculations or decisions happen and what output should be produced.

Order matters: A program must acquire values before using them. Calculating a total before defining item prices is a logic error in the proposed sequence.

Test using an example: Dry-run the steps with example data, record intermediate values and compare the result with what should happen. Correct and refine any unclear step.

Worked example

A program must input two prices, add them and display the total. Write FOUR ordered algorithm steps and explain why the order matters. [4 marks]

1. Input/read the first price [1].

2. Input/read the second price [1].

3. Calculate the sum from the two values [1].

4. Display the total [1]. The calculation must follow both inputs; otherwise required values are unavailable.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A taxi journey fare is a fixed £3 plus £2 per kilometre. Write four ordered steps to calculate and display the fare. [4 marks]

Worked method: 1. Input distance in kilometres [1]. 2. Set base fare = 3 [1]. 3. Calculate fare = 3 + 2 × distance [1]. 4. Output fare [1].

Guided 2

A score program adds a 5-point bonus to an entered score. Explain why the input must happen before calculating the total. [4 marks]

Worked method: The input supplies the original score [1], which is needed for the calculation [1]. The program then adds 5 [1] before displaying the updated result [1].

Guided 3

A pupil writes an algorithm for the area of a rectangle. Explain two criteria that make its steps effective. [4 marks]

Worked method: The steps must be clear and unambiguous [1], so another person can follow them accurately [1]. They must be in the correct order [1], so length/width are known before calculating and displaying area [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the free storage after 7 MB is used.

```
startMB = 20
usedMB = 7
freeMB = startMB - usedMB
print(freeMB)
```

predict

Question 1

State what an algorithm is. [2 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 2

A café calculates the cost of two drinks then adds a £1 service charge. Write FOUR logical steps for the algorithm. [4 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 3

Explain why clear order, unambiguous steps and testing with sample inputs matter when producing an algorithm. [6 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

An algorithm should give a clear sequence of instructions that can be followed to solve a problem.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Breaking a complex problem into smaller manageable subproblems.

Q2. B — Renaming a folder

Q3. Lossless compression restores the original exactly. Losing source-code characters could break the program.

Q4. Utilities maintain or protect a computer, adding tools beyond the OS. Writing an essay is an application task.

I Do / We Do pages · Guided practice

Q1. 1. Input distance in kilometres [1]. 2. Set base fare = 3 [1]. 3. Calculate fare = 3 + 2 × distance [1]. 4. Output fare [1].

Q2. The input supplies the original score [1], which is needed for the calculation [1]. The program then adds 5 [1] before displaying the updated result [1].

Q3. The steps must be clear and unambiguous [1], so another person can follow them accurately [1]. They must be in the correct order [1], so length/width are known before calculating and displaying area [1].

You Do pages · Code First

```
startMB = 20
usedMB = 7
freeMB = startMB - usedMB
print(freeMB)
```

Expected result: 13

You Do pages · Independent questions

Q1. A set/sequence of clear instructions [1] for solving a problem [1].

Q2. Input price of drink 1 [1]; input drink 2 price [1]; total = price1 + price2 + 1 [1]; output total [1].

Q3. Order ensures inputs exist before later calculations [2]. Unambiguous steps specify actions clearly [2], reducing misinterpretation. Testing with sample inputs

[1] allows predicted and actual outcomes to be compared and errors corrected [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: